

CSE 370– Database Systems

Assignment 2

Fall 2024

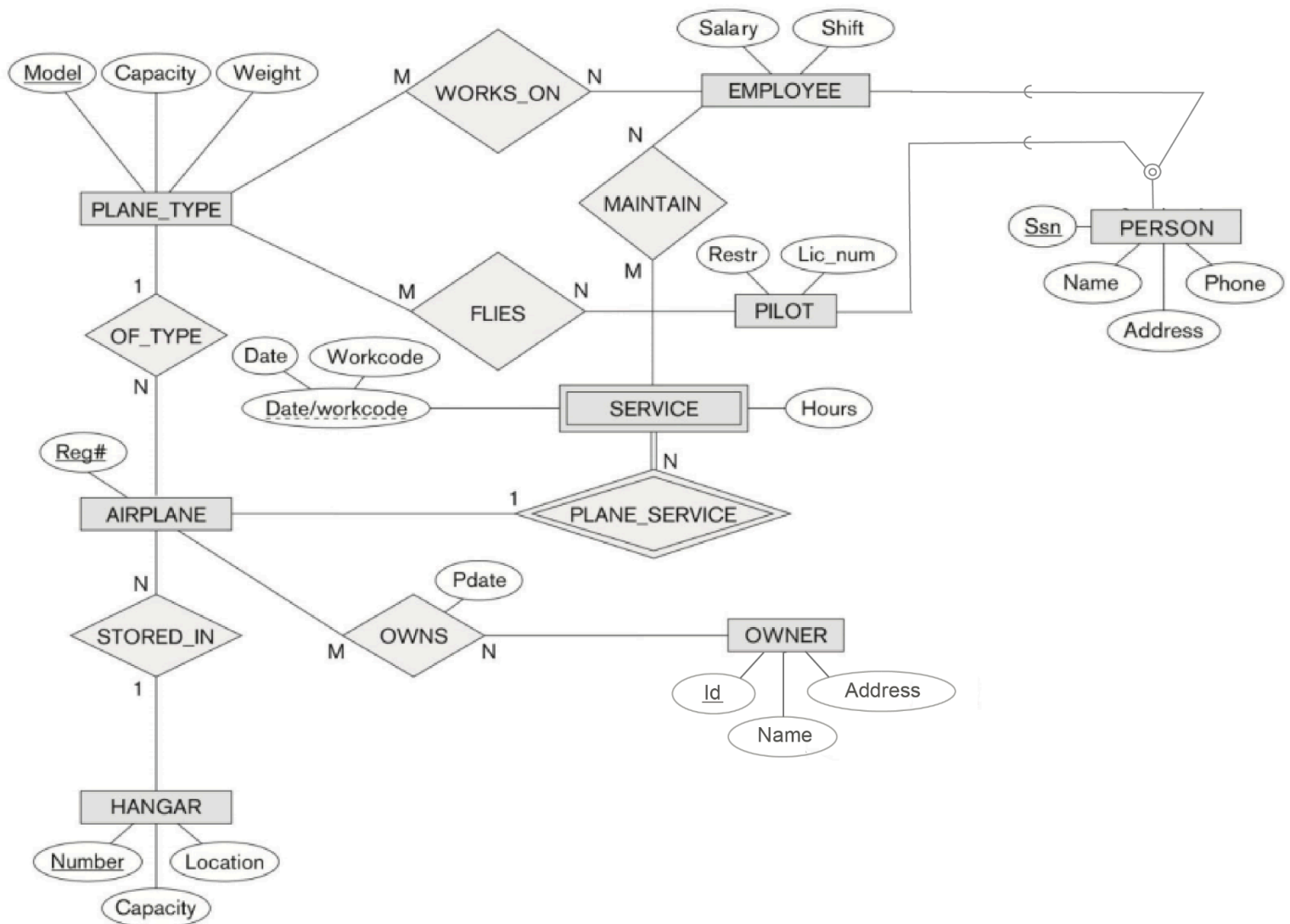
Marks

Full Name (in Block Letter): _____

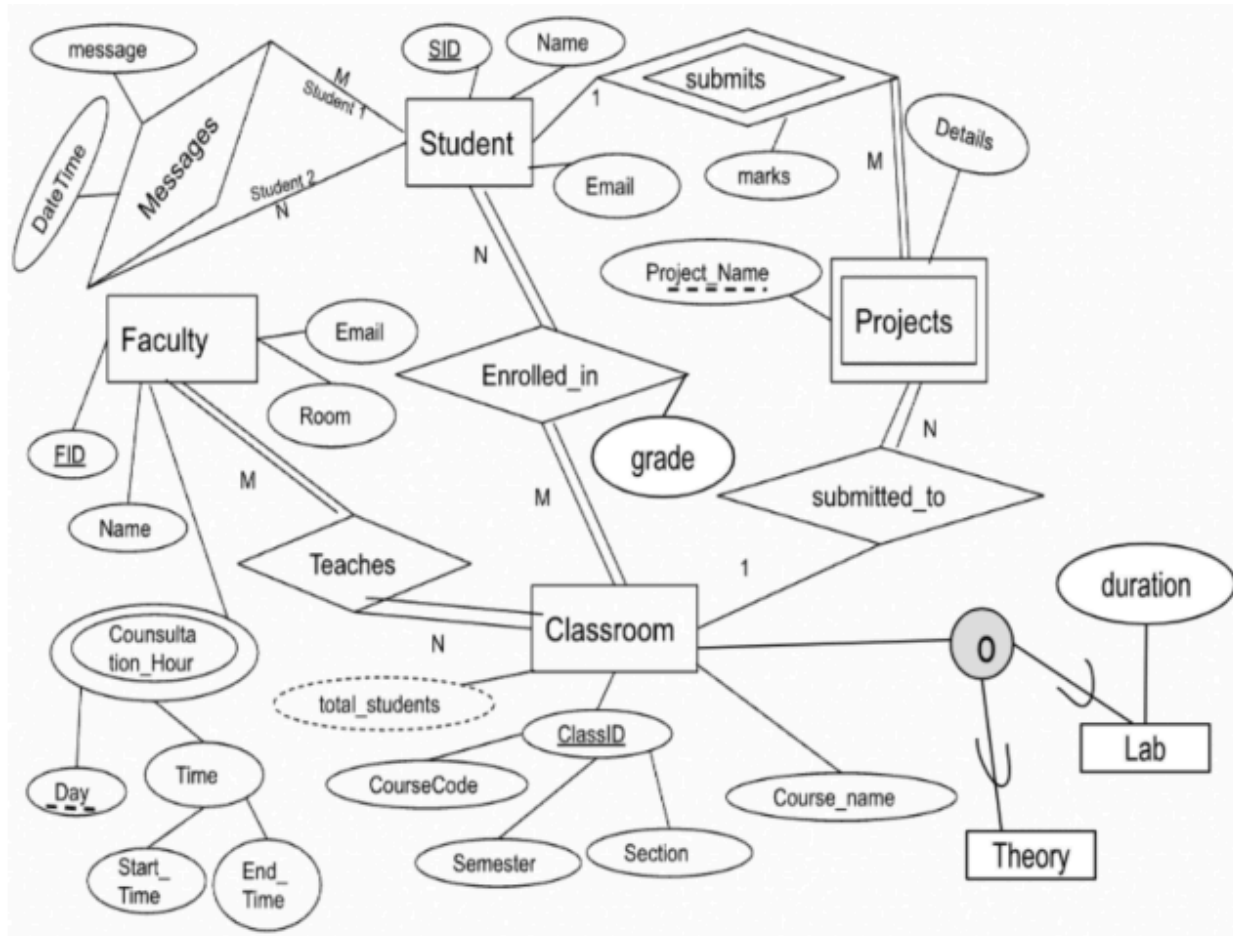
ID: _____ Section: ____ Signature: _____

Date: _____ Total Marks: 10

Question 1 [CO3]: Construct a relational schema diagram from the Extended Entity Relationship (EER) diagram below. Use the best option and give proper justification.

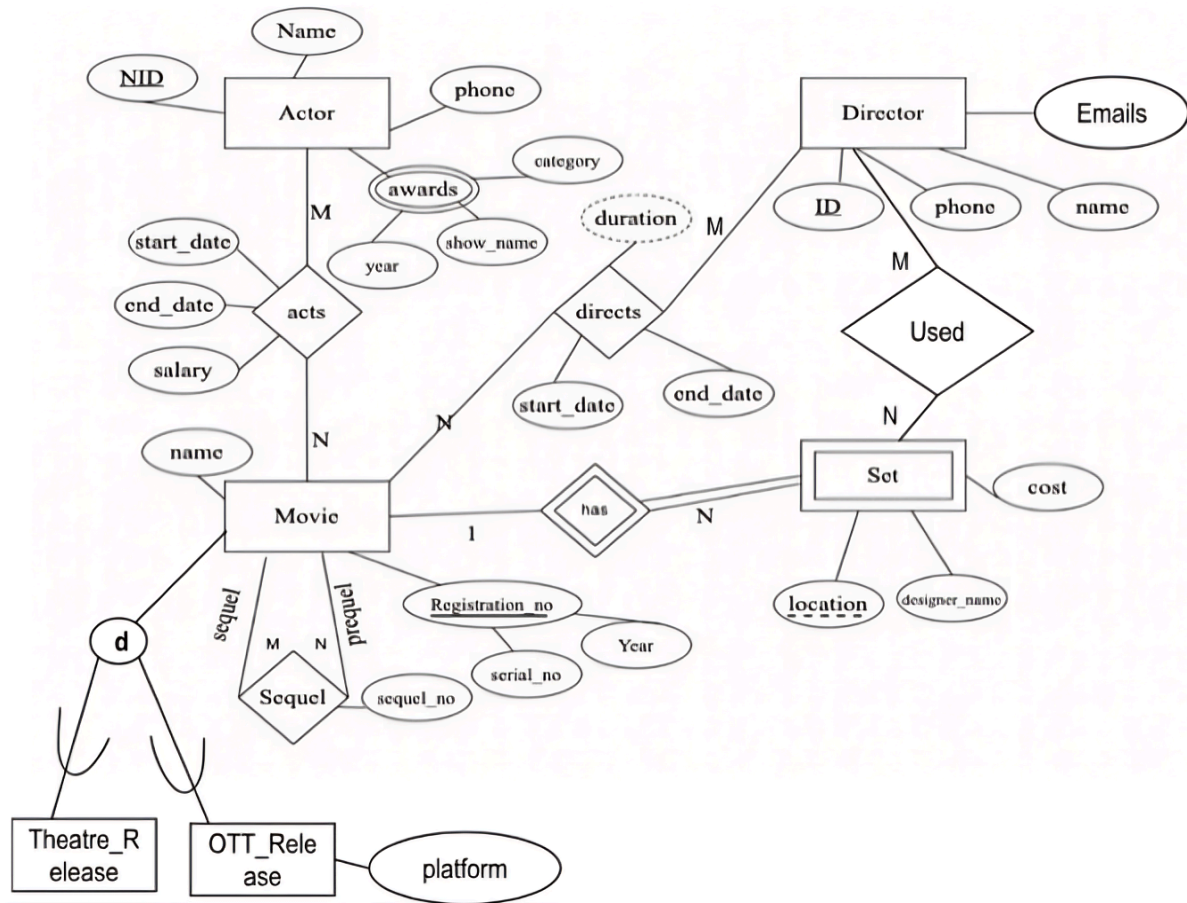


Question 2 [CO3]: Construct a relational schema diagram from the Extended Entity Relationship (EER) diagram below. Use the best option except 8A for the specialization portion. Explain your reasoning for choosing this option.



Question 3 [CO3]:

Construct a relational schema diagram from the Extended Entity Relationship (EER) diagram below. Use the best option except 8A for the specialization portion. Explain your reasoning for choosing this option.



Question 4[CO3]:

Course_Offering (Course_code, Section_no, Semester, Course_title, Department, Credits, Coordinator_initial, Coordinator_name, Coordinator_email, Slot, Room_no, Building_no, Capacity, Total_students)

The primary key of the relation is underlined. The relation has the following additional functional dependencies (FDs):

FD1: Course_code $\rightarrow$ Department, Credits, Course_title

FD2: Course_code, Semester $\rightarrow$ Coordinator_initial, Coordinator_name, Coordinator_email

FD3: Coordinator_initial $\rightarrow$ Coordinator_Name, Coordinator_email

FD4: Building_no, Room_no $\rightarrow$ Capacity

- a. **Explain** if the above relation is in the first normal form (1NF) or not? If not, **apply** 1NF normalization.
- b. **Explain** if the relation(s) of no (a) is/are in the second normal form (2NF) or not? If not, **apply** 2NF normalization.
- c. **Explain** if the relation(s) of no (b) is/are in the third normal form (3NF) or not? If not, **apply** 3NF normalization.

Question 5[CO3]:

Consider the following relation:

Online_Workshop (platformID, workshopID, sessionNo, eventNo, platformName, category, launchYear, headquarters, registrationFee, topicArea, duration, workshopTitle, presenterID, presenterName, presenterEmail, feedbackScore, participantCount)

The primary key is **underlined**.

Relation has the following additional functional dependencies (FDs):

- **FD1:** platformID → platformName, category, launchYear, headquarters, registrationFee
- **FD2:** workshopID → topicArea, duration, workshopTitle
- **FD3:** presenterID → presenterName, presenterEmail
- **FD4:** workshopID, sessionNo, eventNo → presenterID, presenterName, presenterEmail
- **FD5:** category, launchYear → registrationFee

- 1) **Explain** if the above relation is in the first normal form (1NF) or not? If not, **apply** 1NF normalization.
- 2) **Explain** if the relation(s) of no (a) is/are in the second normal form (2NF) or not? If not, **apply** 2NF normalization.
- 3) **Explain** if the relation(s) of no (b) is/are in the third normal form (3NF) or not? If not, **apply** 3NF normalization.